100 Challenges Facing AI Engineers

A comprehensive question bank for interviews, discussions, and self-assessment

1. Data Challenges

1. How do you handle missing or incomplete training data?
2. What strategies do you use to detect and mitigate data bias?
3. How do you ensure data quality at scale?
4. How do you handle imbalanced datasets in classification tasks?
5. What is your approach to data labeling and annotation quality control?
6. How do you manage data versioning across experiments?
7. How do you handle data drift in production systems?
8. What techniques do you use for synthetic data generation?
9. How do you deal with data privacy regulations (GDPR, CCPA) in ML pipelines?
10. How do you validate data pipelines for correctness?
11. What is your strategy for handling duplicate or near-duplicate data?
12. How do you deal with noisy labels in supervised learning?
13. How do you handle multi-source data integration with inconsistent schemas?
14. What approaches do you use for active learning to reduce labeling costs?
15. How do you detect data leakage between training and test sets?

2. Model Development Challenges

16. How do you choose between building a custom model versus using a pre-trained one?
17. What is your approach to hyperparameter tuning at scale?
18. How do you prevent overfitting in deep learning models?
19. How do you debug a model that isn't converging?
20. What strategies do you use for model interpretability and explainability?
21. How do you handle catastrophic forgetting in continual learning?
22. How do you decide on the right model architecture for a given problem?
23. What is your approach to multi-task learning trade-offs?
24. How do you evaluate models when ground truth is ambiguous or subjective?
25. How do you handle reproducibility issues in model training?
26. What techniques do you use for model compression and distillation?

27. How do you approach transfer learning across domains?
28. How do you validate that a model generalizes beyond the training distribution?
29. What is your strategy for handling concept drift?
30. How do you balance model complexity against latency requirements?

3. LLM-Specific Challenges

31. How do you mitigate hallucinations in large language models?
32. What strategies do you use for effective prompt engineering at scale?
33. How do you evaluate the quality of LLM-generated outputs systematically?
34. How do you manage context window limitations for long documents?
35. What is your approach to fine-tuning versus retrieval-augmented generation (RAG)?
36. How do you handle inconsistent outputs from non-deterministic models?
37. How do you reduce latency in LLM inference for real-time applications?
38. What techniques do you use to align LLM outputs with safety guidelines?
39. How do you manage and reduce the cost of LLM API calls at scale?
40. How do you handle multi-turn conversation state and memory management?
41. What is your strategy for chaining multiple LLM calls reliably (agentic workflows)?
42. How do you prevent prompt injection attacks?
43. How do you evaluate and select the right model size for a given task?
44. How do you handle versioning when underlying foundation models change?
45. What is your approach to grounding LLM responses in verified facts?

4. MLOps & Deployment Challenges

46. How do you design a CI/CD pipeline for machine learning models?
47. What is your approach to monitoring model performance in production?
48. How do you handle model rollback when a new deployment underperforms?
49. How do you manage feature stores across training and serving environments?
50. What strategies do you use for A/B testing ML models safely?
51. How do you handle the training-serving skew problem?
52. How do you scale model inference to handle high traffic loads?
53. What is your approach to canary deployments for ML models?
54. How do you manage GPU resource allocation across multiple teams?
55. How do you handle model retraining schedules and triggers?

56. What is your strategy for containerizing and orchestrating ML workloads?
57. How do you ensure low-latency inference for edge deployment?
58. How do you handle multi-region deployment of ML services?
59. What is your approach to dependency management across ML environments?
60. How do you build effective rollback strategies for failed model updates?

5. Ethics, Safety & Governance

61. How do you identify and mitigate algorithmic bias in deployed systems?
62. What is your approach to ensuring fairness across demographic groups?
63. How do you handle the trade-off between model accuracy and explainability?
64. What governance frameworks do you use for responsible AI deployment?
65. How do you handle adversarial attacks against ML models?
66. What is your approach to red-teaming AI systems before launch?
67. How do you ensure compliance with emerging AI regulations?
68. How do you handle user consent and transparency around AI decision-making?
69. What is your strategy for auditing AI systems for unintended consequences?
70. How do you balance innovation speed with responsible AI practices?
71. How do you handle accountability when an AI system causes harm?
72. What is your approach to documenting model limitations for stakeholders?

6. Organizational & Collaboration Challenges

73. How do you communicate model uncertainty to non-technical stakeholders?
74. How do you manage expectations when AI projects have uncertain timelines?
75. What is your approach to cross-functional collaboration with product and design teams?
76. How do you prioritize technical debt versus new feature development in ML systems?
77. How do you onboard new engineers onto complex ML codebases?
78. What is your strategy for knowledge sharing across distributed AI teams?
79. How do you justify AI infrastructure investment to leadership?
80. How do you handle disagreements about model evaluation metrics within a team?
81. What is your approach to managing scope creep in AI projects?
82. How do you balance research exploration with production deliverables?
83. How do you handle the skills gap between research scientists and engineers?
84. What is your strategy for maintaining documentation in fast-moving AI projects?

85. How do you manage stakeholder expectations around AI system reliability?

7. Infrastructure & Scaling Challenges

86. How do you architect systems for distributed model training?
87. What is your approach to managing compute costs for large-scale training runs?
88. How do you handle fault tolerance during long-running training jobs?
89. What strategies do you use for efficient data loading at scale?
90. How do you optimize memory usage for large model training?
91. How do you handle multi-cloud or hybrid-cloud ML infrastructure?
92. What is your approach to caching strategies for repeated inference requests?
93. How do you manage experiment tracking across hundreds of model runs?
94. How do you handle vendor lock-in risk with cloud AI platforms?
95. What is your strategy for capacity planning for unpredictable AI workloads?

8. Future-Facing & Career Challenges

96. How do you stay current with the rapidly evolving AI research landscape?
97. What skills do you think will be most critical for AI engineers in the next five years?
98. How do you evaluate which new AI tools or frameworks are worth adopting?
99. How do you approach career growth as the AI engineering role keeps changing?
100. How do you balance specialization (e.g., LLMs, computer vision) versus generalist skills as an AI engineer?